

The Fisher–KPP Traveling-Wave Atlas for Every Real Speed: Threshold Fronts, Oscillatory Obstructions, and the Route-A Stop

Route-A structural certificate HCS-C202

Abstract

For all diffusion and growth coefficients $D, r > 0$, we classify every real speed in the Fisher–KPP traveling-wave equation. A dimensionless phase-plane triangle proves existence and translation uniqueness of positive monotone fronts exactly when $c \geq 2\sqrt{Dr}$. Spatial reflection closes the negative-speed family; focus tails exclude all nonzero subcritical speeds, while the stationary Hamiltonian has periodic ovals but no requested heteroclinic. We derive the subcritical, supercritical and repeated-root asymptotics and exclude nonconstant cycles for $c \neq 0$.

Keywords: Fisher–KPP equation; traveling front; minimal speed; phase plane; heteroclinic orbit; reaction–diffusion.

中文摘要

对任意扩散系数与增长率 $D, r > 0$ ，本文统一分类 Fisher–KPP 行波方程的全部实速度。相平面不变三角形证明正单调波前恰在 $c \geq 2\sqrt{Dr}$ 时存在，且除平移外唯一。负速度由空间反射得到；亚临界速度的焦点尾部必然变号， $c = 0$ 虽有哈密顿周期卵形，却不存在所需异宿轨道。三种临界尾部与精确特殊解共同锁定全部符号和尺度；有限计算只作回归，不替代连续存在性证明。该模型没有内生素数载体、周期轨道层或目标谱结构，故严格否决路线 A。
关键词：Fisher–KPP 方程；行波；最小速度；相图；异宿轨道。

1 Frozen equation and exact scaling

Fix $D, r > 0$ in physical time and consider

$$u_t = Du_{xx} + ru(1 - u). \quad (1)$$

Fisher introduced this logistic diffusion model and its wave argument [1]; Kolmogorov, Petrovskii and Piskunov established the classical minimal-speed front theory [2]. With $z = x - ct$, $u(x, t) = U(z)$, (1) becomes

$$DU_{zz} + cU_z + rU(1 - U) = 0. \quad (2)$$

Set

$$\xi = \sqrt{r/D} z, \quad s = c/\sqrt{Dr}, \quad V = U_\xi.$$

The complete parameter family is conjugate, with time orientation retained, to

$$U_\xi = V, \quad V_\xi = -sV - U(1 - U). \quad (3)$$

The equilibria are $(1, 0)$ and $(0, 0)$. The first is always a saddle; the zero-state characteristic polynomial is

$$\lambda^2 + s\lambda + 1. \quad (4)$$

2 The positive-speed front theorem

Assume $s \geq 2$ and define the slow positive slope

$$q = \frac{s - \sqrt{s^2 - 4}}{2}, \quad q^2 - sq + 1 = 0.$$

Consider the compact triangle

$$\mathcal{T}_q = \{(U, V) : 0 \leq U \leq 1, -qU \leq V \leq 0\}. \quad (5)$$

On $V = 0$, $V_\xi = -U(1 - U) < 0$, while on $U = 1$ one has $U_\xi = V \leq 0$. On the slanted boundary put $G = V + qU$; substitution of $V = -qU$ gives the decisive identity

$$G_\xi = U\{q(s - q) - 1 + U\} = U^2 > 0. \quad (6)$$

The decreasing branch of the one-dimensional unstable manifold of $(1, 0)$ enters $\mathcal{T}_q$. Since the divergence of (3) is the negative constant $-s$, a nonconstant periodic omega limit is impossible. The energy identity below also prevents return to the saddle; hence the branch converges to $(0, 0)$. It is a profile with

$$0 < U < 1, \quad U_\xi < 0, \quad U(-\infty) = 1, \quad U(+\infty) = 0. \quad (7)$$

Conversely, every orbit with (7) must leave the saddle along this same unstable branch. Reparametrizing one orbit changes only the origin of ξ . Thus for every $c \geq 2\sqrt{Dr}$ the front exists and is unique modulo translation.

3 Every remaining speed

If $W(\xi) = U(-\xi)$, then W solves (3) with speed $-s$. Therefore every $c \leq -2\sqrt{Dr}$ has a unique reflected positive profile, increasing from 0 to 1.

When $0 < s < 2$, the roots of (4) are

$$-\frac{s}{2} \pm i \frac{\sqrt{4-s^2}}{2}.$$

Asymptotic linearization at this hyperbolic focus makes every nontrivial tail approaching zero rotate and change the sign of U . It cannot be a $[0, 1]$ front. Reflection gives the same obstruction for $-2 < s < 0$.

At $s = 0$, system (3) conserves

$$H_0(U, V) = \frac{V^2}{2} + \frac{U^2}{2} - \frac{U^3}{3}. \quad (8)$$

For every $0 < h < 1/6$, the bounded component of $H_0 = h$ is a periodic oval around the center $(0, 0)$; it crosses negative U and is not a positive front. More decisively, $H_0(1, 0) = 1/6 \neq 0 = H_0(0, 0)$, so no stationary 1-to-0 heteroclinic exists. This closes the full real speed axis.

4 Energy, no cycles, and three leading-edge laws

In physical variables,

$$E = \frac{D}{2}U_z^2 + r\left(\frac{U^2}{2} - \frac{U^3}{3}\right), \quad E_z = -cU_z^2. \quad (9)$$

Consequently no nonconstant periodic profile exists for $c \neq 0$; this is also Bendixson's conclusion from divergence $-c/D$.

At the back of a positive front,

$$1 - U(z) \sim Ce^{\alpha z}, \quad C > 0, \quad \alpha = \frac{\sqrt{c^2 + 4Dr} - c}{2D} > 0, \quad z \rightarrow -\infty.$$

At the leading edge $z \rightarrow +\infty$, three distinct laws occur:

$$c > 2\sqrt{Dr}: \quad U(z) \sim Be^{\lambda_+ z}, \quad B > 0, \quad \lambda_+ = \frac{-c + \sqrt{c^2 + 4Dr}}{2D}. \quad (10)$$

$$c = 2\sqrt{Dr}: \quad U(z) \sim (Az + B)e^{-\sqrt{r/D}z}, \quad A > 0. \quad (11)$$

$$0 < c < 2\sqrt{Dr}: \quad U(z) = e^{-cz/(2D)}\{A \cos(\omega z) + B \sin(\omega z) + o(1)\}, \quad (12)$$

$$\omega = \frac{\sqrt{4Dr - c^2}}{2D}.$$

Here $(A, B) \neq (0, 0)$ for a nontrivial tail. For (10), the triangle excludes the fast eigendirection. At the critical speed, $W = e^{\sqrt{r/D}z}U$ obeys $W'' = (r/D)e^{\sqrt{r/D}z}U^2 > 0$, giving the positive generalized-eigenvector coefficient A . Equation (12) is the announced sign-changing obstruction, not a numerical shooting observation.

Revision focus. Round 1 adds reflection, every forbidden speed, Hamiltonian ovals, energy/no-cycle identities and all three tail regimes.

Declarations

Data and code availability. Package-local deterministic code and synthetic exact evidence accompany this manuscript.

Ethics. No human, animal, clinical, personal or sensitive data are used; approval is not applicable. Historical sources are cited for mathematical provenance, not ideological endorsement.

Funding and conflicts. No external funding is reported; no conflict of interest is known.

Author contributions (CRediT). This anonymous certificate records Conceptualization, Formal analysis, Software, Validation, Writing—original draft, and Writing—review and editing. AI systems are not authors.

AI-use disclosure. An AI coding assistant supported drafting and exact-code development; it was not an external reviewer or independent error process.

References

- [1] R. A. Fisher, "The Wave of Advance of Advantageous Genes," *Annals of Eugenics* 7(4) (1937), 355–369. DOI: 10.1111/j.1469-1809.1937.tb02153.x.
- [2] A. N. Kolmogorov, I. G. Petrovskii and N. S. Piskunov, "Étude de l'équation de la diffusion avec croissance de la quantité de matière et son application à un problème biologique," *Bull. Univ. État Moscou, Série Int. A* 1 (1937), 1–25. English translation in *Selected Works of A. N. Kolmogorov I*, Kluwer, 1991, 242–270.
- [3] M. J. Ablowitz and A. Zeppetella, "Explicit solutions of Fisher's equation for a special wave speed," *Bull. Math. Biol.* 41 (1979), 835–840. DOI: 10.1007/BF02462380.